



Semi-autogenous mill power model development using gene expression programming

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ABSTRACT

Predicting the performance of the semi-autogenous (SAG) mill is necessary for the best circuit design which is possible by suitable modeling and simulation. Numerous models of the SAG mill are studied in the literature, but the majority of them do not evaluate the predicted model for full-scale mill performance. Mill powers of the semi-autogenous mill have an effective influence on the mill performance. In this regard, a new predictive model based on gene expression programming (GEP) was developed to predict the mill power of the SAG mill. To achieve this purpose, a total number of 186 full-scale SAG mill works were investigated and the most effective parameters on SAG mill power, i.e., feed moisture, mass flowrate, mill load cell weight, SAG mill solid percentage, inlet and outlet water to the SAG mill and work index were measured and utilized to develop the GEP model. In order to determine the relationship between the input and output parameters, the GEP model was developed and the results were compared with non-linear multiple regression (NLMR) method. The results show the capability of the GEP model in predicting the mill power. It shows that the mill power is more sensitive to mass flowrate and work index than other input parameters.

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1. Introduction

Semi-autogenous grinding (SAG) has been recently selected as the primary stage of grinding due to the economic considerations in most mining companies. SAG mills have some advantages such as lower physical space requirements, lower investment and maintenance costs and higher processing capacity, compared to the conventional circuits [1]. However, it has greater complexity in operation and control because of its large pieces of equipment. Semi-autogenous mill modeling is difficult because of the interaction between affecting parameters during the process such as mill performance and feed characteristics and inaccessibility of appropriate full-scale circuit data [2]. The Bond approach is based on the size reduction which follows as in Eq. (1):

$$W = W_i \left(\frac{10}{\sqrt{P}} - \frac{10}{\sqrt{F}} \right) \quad (1)$$

where W , W_i , P and F are the specific energy, the work index, the 80% passing size for the product and the 80% passing size for the feed, respectively.

The specific energy of a circuit is higher than what this equation predicts; thus, it is assumed that this equation has less efficiency. Bond's equation with a high applicability is used for the comparison of different circuit designs and specially the use of so-called "operating work index" (OW_i). The operating work index associated with Bond's equation is written as [3]:

$$OW_i = \frac{W}{10} \left(\frac{1}{\sqrt{P}} - \frac{1}{\sqrt{F}} \right) \quad (2)$$

The comparison of different circuits with different feed and product sizes can be carried out using the operating work index in the theory [3]. The refinement of the Bond models and more studies of mill charge detailed characterization enhanced the development of several mill powerdraw models [4,5]. Eq. (3) was developed by the Julius Kruttschnitt Mineral Research Centre as:

$$P_{Gross,JK} = P_{No Load} + kP_{Charge} \quad (3)$$

where, $P_{Gross,JK}$, $P_{No Load}$, P_{Charge} and k are the mill powerdraw (kW), the empty mill powerdraw (kW), the powerdraw of the entire contents of the mill (kW) and a lumped mill powerdraw parameter, respectively [6].

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The JA Herbst and Associates also developed the powerdraw model (Eq. (4)):

$$P_{Gross,JAH} = C_3 \sin(\alpha) D_m^{0.3} W_C (3.2 - 3V^*) N^* \left(1 - \frac{0.1}{2^{9-10N^*}} \right) \quad (4)$$

where $P_{Gross,JAH}$ is the mill powerdraw (kW), N^* is the fractional critical speed, W_C is the charge mass (t), V^* is the mill fraction occupied by the charge, α is the charge angle of repose ($^\circ$); and C_3 is a constant [7].

Apelt et al. combined the mill powerdraw, weight models and the related plant measurement to study the indirect measurement of the mill inventories. They showed that the investigation from the weight based models has the least uncertainty [7]. Morrell also predicted the specific energy of the AG and SAG mill using small diameter drill core samples. They developed a new rock breakage characterization test to generate a strength index that can be used for the specific energy of AG and SAG mills [2]. Morrell presented a new method for predicting the specific energy requirement of tumbling mill (grinding) circuits. They showed that the energy utilization efficiency of all studied plant grinding circuits had no significant difference [3]. Salazar et al. presented the dynamic modeling and simulation of semi-autogenous mills for predicting the time-evolution of product flow rate, level charge, powerdraw and load position using the conventional non-stationary population balance approach. This dynamic simulator can be used for milling operations in order to design and evaluate the advanced control schemes [1]. Completed models can be simulated with the lower risk of not piloting than pilot-scale test works based on a range of full-scale circuits [2].

This work investigates the use of feed moisture, mass flowrate, mill load cell weight, SAG mill solid percentage, inlet and outlet water to the SAG mill and work index to obtain the estimated mill power. The SAG mill circuits can be optimized using the modeling and simulation that can predict the throughput, power draw and product size distribution. For this aim, a new predictive model based on GEP is suggested to estimate the SAG mill power. In order to show the capability of the constructed model in predicting SAG mill power, non-linear multiple regression was performed. Finally, the performance predictions of the developed models were compared and discussed.

2. Process description and data analysis

Statistical analysis of the dataset is performed based on the 186 SAG mill operations collected from Aq Darreh gold processing plant 32 km north of Takab city in West Azarbayjan province, Iran. The plant is

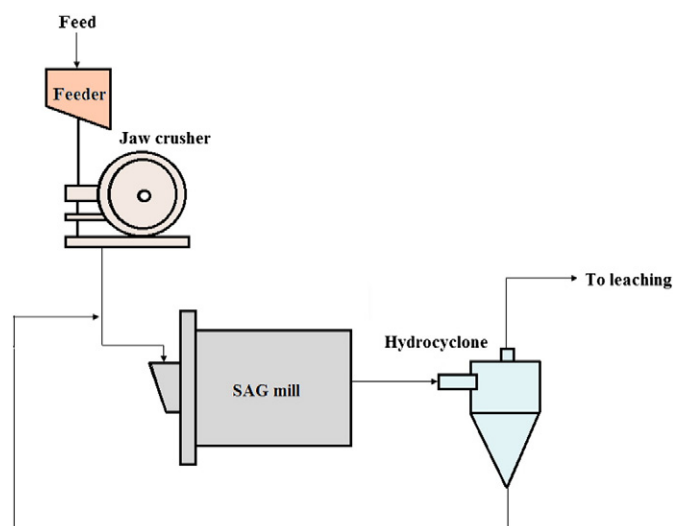


Fig. 1. Primary grinding circuit process flowsheet.

supplied from the Aq Darreh mine, at a distance of 12 km. The ore contains about 3 ppm gold. Fig. 1 shows the grinding flowsheet of Aq Darreh gold processing. Ore is fed to the jaw crusher for crushing after passing from the grizzly screen. The crusher discharge is fed to the SAG mill for grinding and mill discharge is classified by hydrocyclones. The over flow of hydrocyclones is fed for leaching in order to recover the gold and silver from the ore. The variables of feed moisture [M] (%), mass flowrate [TPH] (t/h), mill load cell weight [LC] (t), SAG mill solid percentage [S] (%), inlet [IW] (m³/h) and outlet water [OW] (m³/h) to the SAG mill and work index [WI] (kWh/t) are measured for modeling the SAG mill power [KW] (kW).

2.1. Data analysis

In datasets, the outlying data points can have an adverse influence on the relationships clear understanding among the variables. In order to make a predictable model of the SAG mill power, the outlying data points are identified from the dataset to be eliminated for building a more homogeneous dataset using multivariate statistical tools of the principal components and factor analyses. The basic descriptive statistical analysis of the original dataset is shown in Table 1. The original dataset box plot is shown in Fig. 2. As Fig. 2 shows, the median of most data groups is not in the box center, which indicates their distribution is not symmetric. The variables of TPH, S and IW do not have any outliers whereas M, OW, LC and WI have at least one outlier.

In order to detect the outliers and natural groups of data, the multivariate statistical tool of the principal components analysis is also used [8,9]. It can be used for the dimension reduction for the problems with high dimensions. Principal components in the PCA are the vectors which explain the most variance of the dataset that is a linear combination of original variables. The dataset correlation matrix is obtained by using the principal components analysis [10].

Table 2 shows the coefficients of the principal components. The percentage of variability explained for each principal component is shown in Fig. 3 using the Pareto chart that shows the relative importance of the differences between groups of data. It is an appropriate way for recognizing the causes of quality problems or loss. It can be also used for deciding the data group with the most attention [9]. The decrease in component variance is shown with the columns in Fig. 3. As Fig. 3 shows, the highest component variance is the first principle component. All variables in the first principal component and LC, WI and KW in the second principal component have the absolute largest coefficients. In the third component, M, LC, S, IW and KW are mainly weighted. The scatter plot of the second principal component (PC2) versus the first principal component (PC1) (Fig. 4) shows that there are no natural groups in the dataset whereas the outlying data points (they were determined with a red circle) can be seen in the dataset. The outlier data (44 data) were removed from the original data to build a more homogeneous dataset that was used for the modeling.

Table 1
Basic descriptive statistics for the original database.

Variable	Unit	Minimum	Maximum	Mean	Std. deviation
Moisture (M)	%	8.800	17.850	11.898	1.694
Mass flowrate (TPH)	t/h	55.218	155.792	110.827	22.182
mill load cell weight (LC)	t	61.040	142.100	98.116	12.486
SAG mill solid percent (S)	%	59.300	68.000	64.815	2.085
Inlet water (IW)	m ³ /h	17.830	74.300	51.304	13.768
Output water (OW)	m ³ /h	10.400	115.5	85.520	19.816
Work index (WI)	kWh/ton	10.488	24.340	15.001	2.713
Mill power (KW)	kW	1225	1798	1607.488	135.818

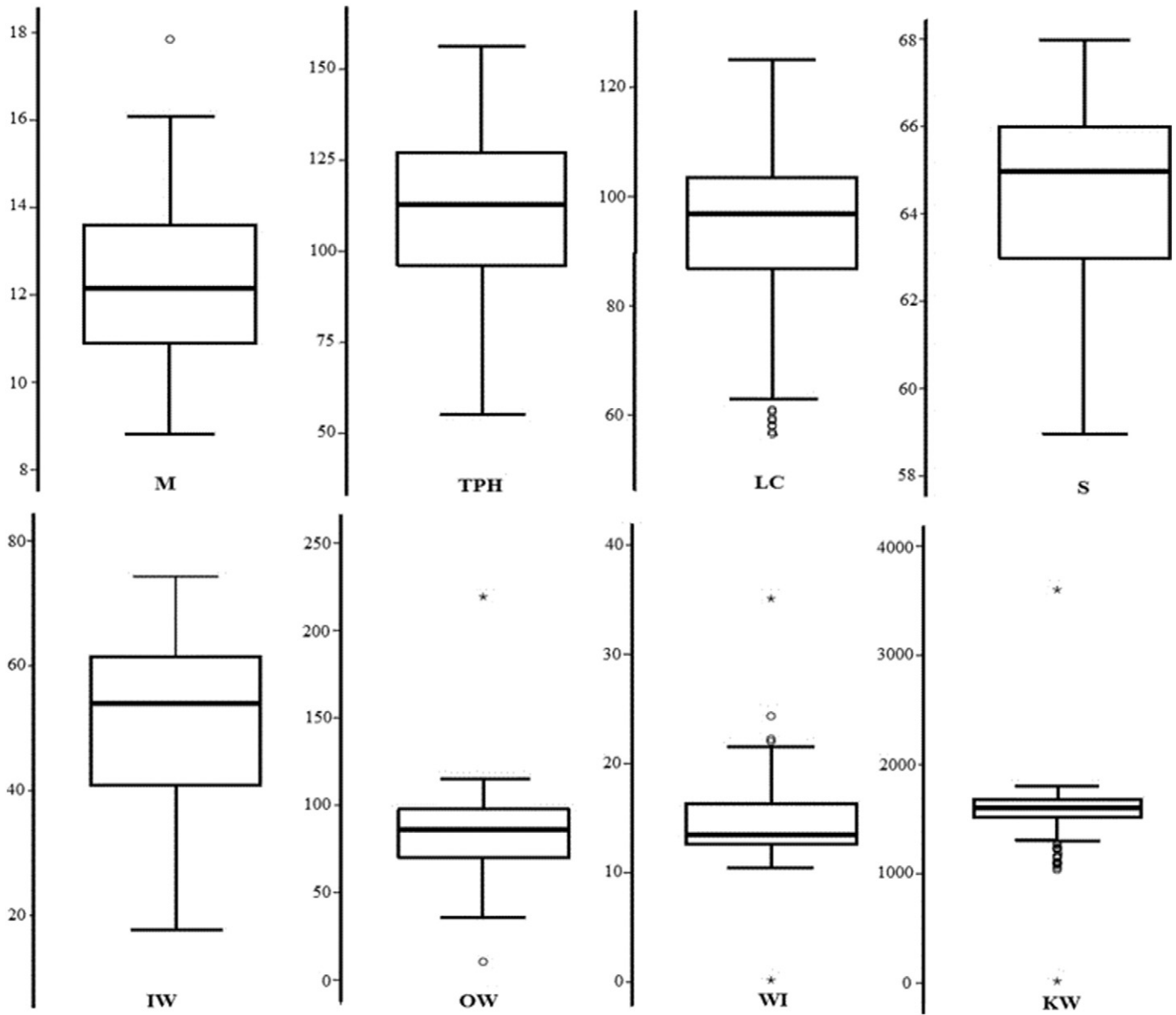


Fig. 2. Box plot of the original dataset.

Table 2
Coefficients for the PCs.

Variables	Principal components							
	PC1	PC2	PC3	PC4	PC5	PC6	PC7	PC8
M	−0.301	−0.152	0.776	−0.260	0.455	0.072	−0.050	0.036
TPH	0.439	−0.196	0.181	−0.098	−0.190	−0.282	−0.371	0.688
LC	0.330	0.312	0.293	0.787	0.289	−0.066	0.028	−0.018
S	0.365	0.160	−0.389	−0.344	0.752	0.004	−0.064	0.041
IW	0.435	−0.160	0.204	−0.214	−0.095	−0.287	0.766	−0.139
OW	0.406	−0.110	0.119	−0.047	−0.137	0.888	0.018	0.007
WI	−0.236	0.660	0.022	−0.131	−0.074	0.156	0.380	0.563
KW	0.254	0.585	0.267	−0.346	−0.273	−0.130	−0.351	−0.433
Eigenvalue	4.452	1.660	0.754	0.417	0.353	0.291	0.062	0.010
Variability (%)	55.655	20.756	9.428	5.219	4.410	3.637	0.775	0.119
Cumulative (%)	55.655	76.412	85.840	91.059	95.469	99.106	99.881	100.000

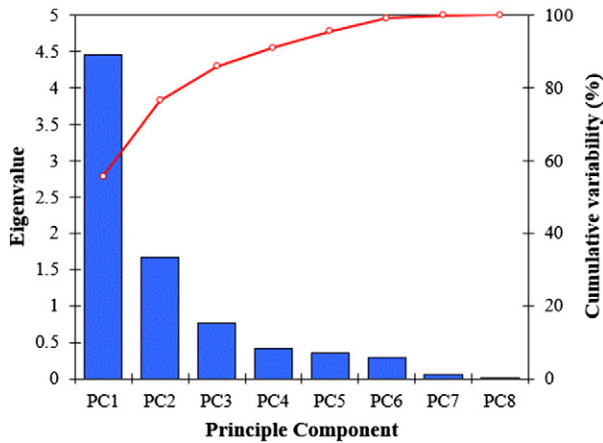


Fig. 3. Scree plot for the first six PCs.

2.2. Effect of parameters

The results showed that the mill power has an inverse relation with the feed moisture, and a direct relation with work index, inlet and outlet water to the SAG mill, SAG mill solid percentage, mill load cell weight and mass flowrate. One of the important factors in the SAG mill power is the work index. The feed hardness influences on SAG mill operation is more significant than other mill operations such as rod mill and ball mill. The feed hardness affects both the power draw and the grinding performance of the mills. The variation in size and hardness of feed led to changes in the mill mass flowrate [11]. Fig. 5 shows that the mill power increases with increasing the work index and mass flowrate. The mill power increases with increasing inlet water to the SAG mill, SAG mill solid percentage and mass flowrate due to increasing mill load weight.

3. Symbolic regression using gene expression programming

Ferreira (2001) proposed a new population-based evolutionary algorithm to overcome the shortages of GA and GP techniques called gene expression programming (GEP) [12,13]. GEP is an evolutionary algorithm which is able to introduce a mathematical equation between

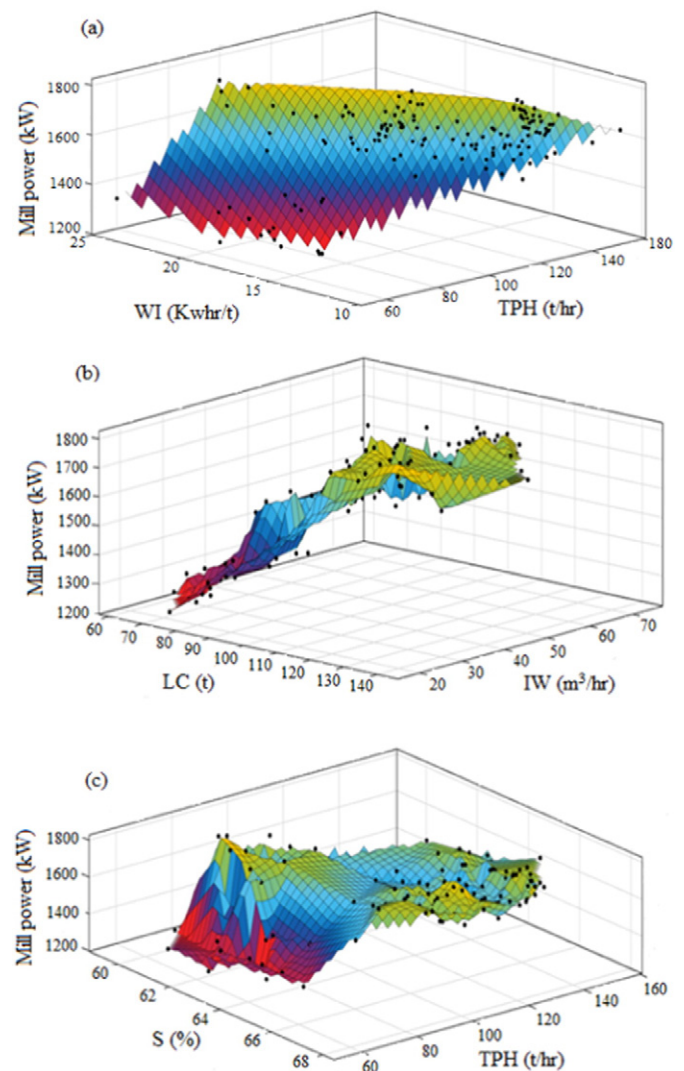


Fig. 5. The effect of input parameters on the mill power.

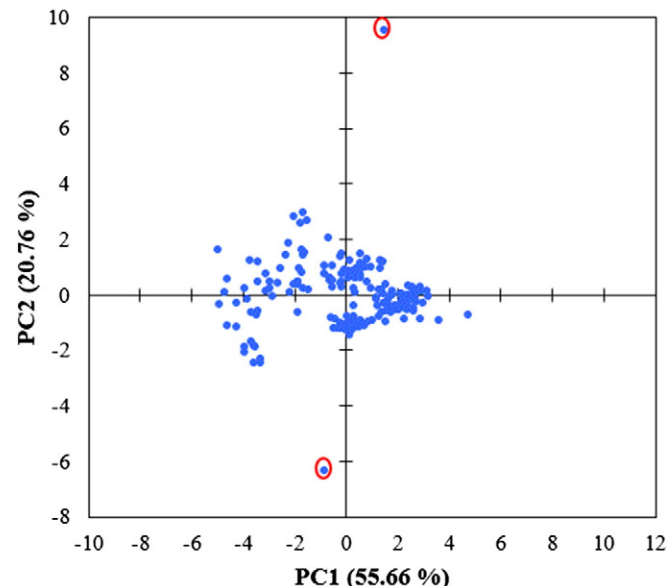


Fig. 4. Scatter plot of second PC versus the first one (before removing outliers).

input parameters to predict output with high accuracy and low estimation error. This algorithm can check the entire search space and find the global optimum without getting stuck in local minimum. Application of the GEP method in mining engineering is limited in some studies in blasting and rock mechanics fields. In the blasting field, several studies have been conducted to predict ground vibration [14–16] and flyrock [17,18] due to the blasting operation in open pit mines. In these studies, the performance prediction of the developed GEP model was compared with ANN and non-linear multiple regression (NLMR) methods and the results showed the excellence of the GEP model. The literature review showed the prosperous application of the GEP algorithm to estimation of uniaxial compressive strength (UCS) of various rock types [19], settlement due to tunneling [20] and air flow rate in single rock joints [21].

GEP enjoys the simplicity of the genetic algorithm and the capabilities of genetic programming [22,23]. The main elements of the GEP algorithm are the individuals (solutions) which are fixed-length string characters (chromosomes) like GA that follows the Karva language (Karva is a language that was invented by Ferreira (2001) to read the expression of the coded programs in the chromosomes). These coded solutions have a tree-like structure called expression tree (ET). A chromosome in GEP may consist of one or more genes. Each gene is comprised of two sections: head and tail (see Fig. 6). The head contains symbols that represent both functions and terminals, whereas the tail contains only the terminals. Terminals are the inputs of the model (input variables and constant values) and functions are the basic

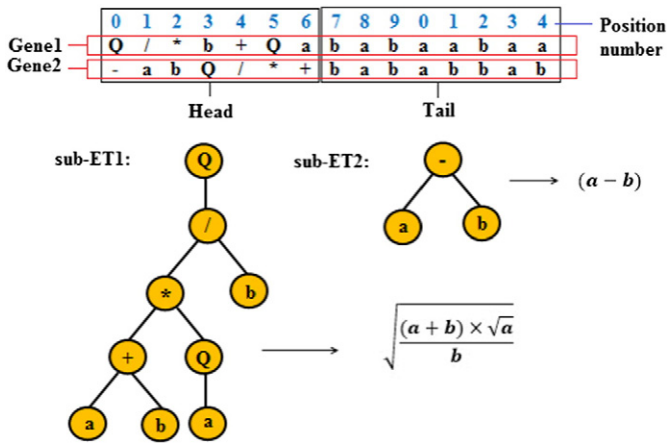


Fig. 6. The Karva expression of two genetic chromosomes with its ETs.

arithmetic functions (e.g. $\{+, -, *, /\}$), nonlinear functions (e.g., $\{\sin, -\cos, \tan, \arctan, \sqrt{}, \sqrt[3]{}, \text{and } \exp\}$) and/or Boolean operators (e.g. $\{\text{Not}, \text{Nor}, \text{Or}, \text{And}\}$). It should be mentioned that the functions of $\sin, -\cos, \tan, \arctan, \text{and } \sqrt{}, \sqrt[3]{}$ and $\exp$ represent the sine, cosine, tangent, arctangent, square root, third root and the exponential of a variable. For each problem, the length of head (h) is chosen by the

GEP designer but the length of tail (t) is a function of h and the maximum number of argument ($n_{\max}$) is related to functions and evaluated by the following equation:

$$t = h(n_{\max} - 1) + 1 \quad (5)$$

In GEP, chromosomes are made from the random combination of terminals and functions. Each one of the genes represent a sub-ET that will be finally linked to each other by the linking function (e.g. $+, -, *, /$) [15,16,21,24].

According to Fig. 7, the GEP procedure begins with the random generation of chromosomes in the form of the Karva language (K-Expression) as the first population. Afterwards, each one of the chromosomes is expressed as ET and mathematical equations, respectively. Based on the defined fitness function, the cost of each individual is evaluated. The individuals are then selected by the roulette wheel (fitness-proportionate) selection method [25] to form the next generation. Suppose that $f(s_i(t))$ is the fitness of the individual s_i in the population at generation t . Then, under fitness-proportionate selection, the probability of individual s_i will be copied into the next generation of the population as a result of any one reproduction operation as [26]:

$$\frac{f(s_i(t))}{\sum_{j=1}^M f(s_j(t))} \quad (6)$$

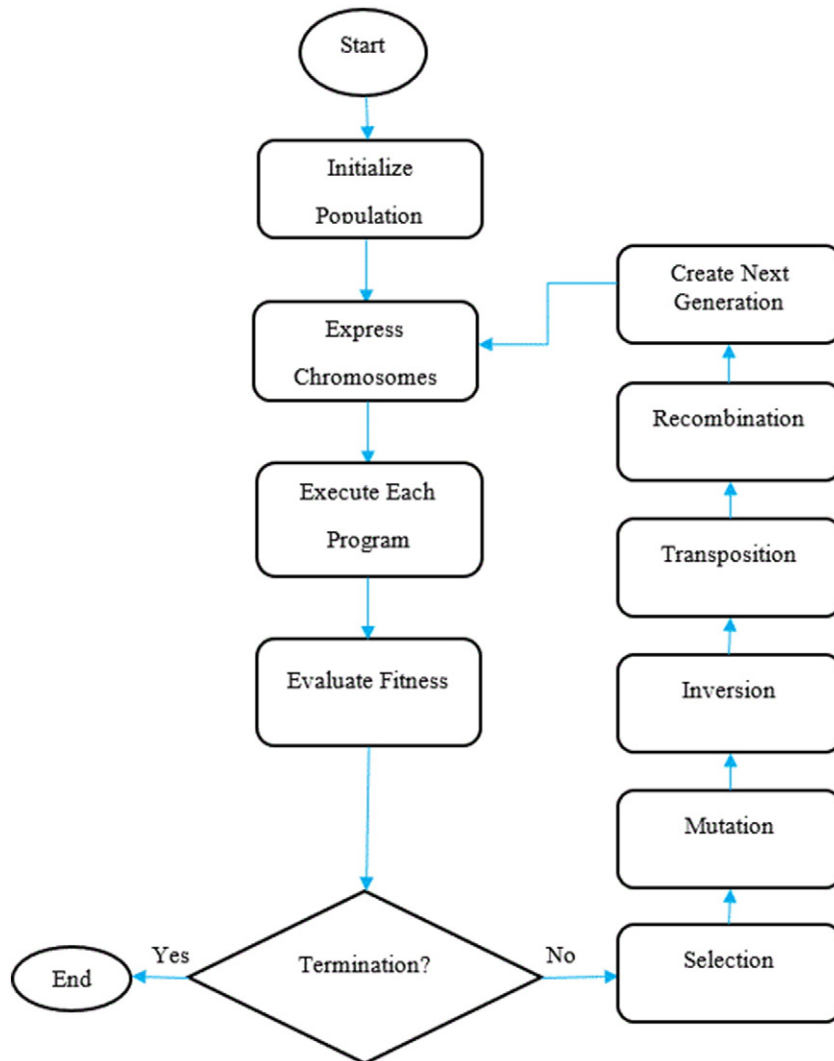


Fig. 7. Flowchart of GEP algorithm.

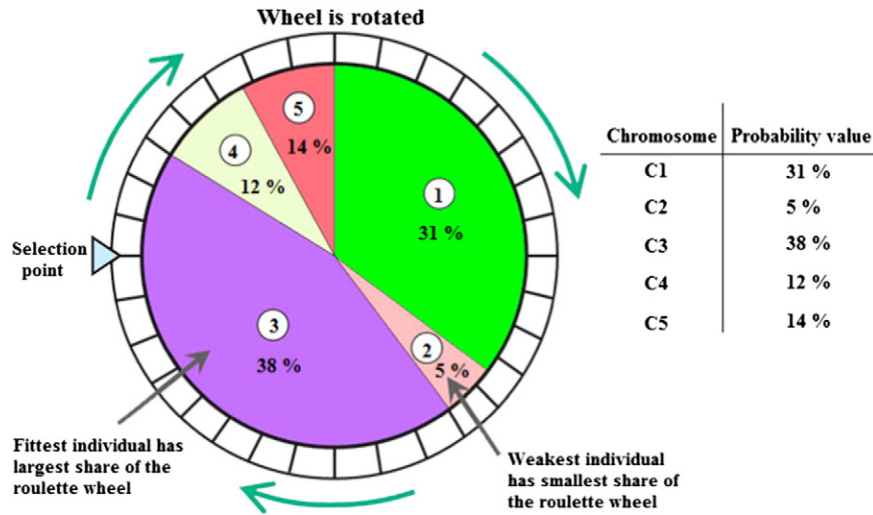


Fig. 8. Roulette wheel selection method.

where, $\sum_{j=1}^M f(s_j(t))$ is the total amount of fitness values with M number of chromosomes. In the following, the probability values of chromosomes are calculated as a sector of the circle and placed on it and the process of selection will be done by the selector (see Fig. 8). The individuals with higher fitness have more chances for the selection.

Then, the genetic operators including mutation, inversion, transposition and recombination are applied to the chromosomes in order to create the modified individuals for the next generation. In a recursive path, this process is repeated for special numbers of generations until the appropriate solution is found. Genetic operators have been introduced in the following [19,27–29]:

3.1. Mutation

Mutation is the most efficient operator with intrinsic modification power which can occur anywhere in the chromosome length. In the head, mutation can change any function or terminal to each other whilst in the tail, it can only convert a terminal to another terminal. Scholars suggest the range of 0.01 to 0.1 for this operator [12,26,30].

3.2. Inversion

The activity domain for this operator is limited in the head of chromosomes. Inversion reverses a fragment in the head usually with the length of 1 to 3. The suggested ratio for this operator is 0.1.

3.3. Transposition

There are three kinds of transposition operators including: 1) Insertion sequence transposition (IS transposition): a fragment with function or terminal in the first position transposes to the head of its own or other genes; 2) Root insertion sequence (RIS) transposition: a segment with a function in the first position transposes the root of the genes; 3) Gene transposition: This operator transposes all genes to the beginning of the chromosome. Scholars suggest the range of 0.01 to 0.1 for this operator [12,26,30].

More details about the genetic operators can be found in the literature [23,26,28,30,31].

4. Results and discussion

Prediction of the SAG mill power is evaluated using NLMR and GEP. The performance of these methods are compared with each other.

4.1. Mill power model development using NLMR

In order to identify the relationship between the input and output variables systematically in the form of a function, a regression analysis can be done as a suitable statistical tool. The multiple regression analysis is a technique which is used to fit a model into the data observations when there are several independent variables. NLMR is a regression in which the dependent or criterion variables are modeled as a non-linear function of model parameters and one or more independent variables. There are several types of NLMR models in the form of exponential, power and logarithmic [16,21,32]. The same training and testing datasets with the GEP model were used for NLMR modeling. To develop the NLMR model, version 21 of the SPSS software package was employed. While developing these models, M , TPH , LC , S , IW , OW and WI were considered as the input variables. Finally, the equations obtained by the NLMR technique are shown in Eq. (7). Performance and accuracy of the constructed GEP and NLMR models have been investigated in Section 4.3.

$$\text{Mill power} = \frac{500 * (TPH^{0.650}) + 500 * LC + 109.315 * \exp(0.080 * S) + 99.999 * \exp(-10.437 * IW) + 60.0958 * \exp(M^{-1.369}) + 108.869 * (OW^{0.15404}) + 109.623 * (WI)}{60.0958 * \exp(M^{-1.369}) + 310} \quad (7)$$

Table 3

Parameters of the developed GEP model.

Description of parameter	Setting of parameter
Fitness function	RMSE
Number of chromosomes	100
Number of generation	2000
Head size	14
Number of genes	5
Linking function	Addition
Mutation rate	0.04
Inversion	0.2
IS transposition rate	0.2
RIS transposition rate	0.2
Gene transposition rate	0.1
One-point recombination	0.3
Two-point recombination	0.3
Gene-recombination	0.1

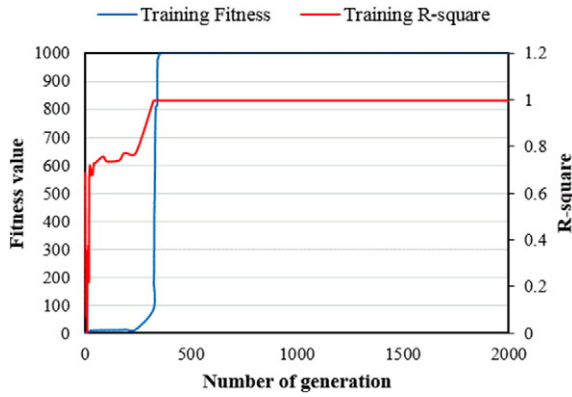


Fig. 9. Variation of fitness value and R^2 with generation number in the training step.

4.2. Mill power model development using GEP

The main objective of this section is developing the mathematical equation based on the GEP method in the form of Mill power = $f(M, TPH, LC, S, IW, OW, WI)$ where M , TPH , LC , S , IW , OW and WI are the model inputs (independent variables) and mill power is the output of the model (dependent variable). For this purpose, a powerful software namely GeneXproTools 4.0 was used [33].

First of all, all of the 142 datasets were separated randomly to two sections of training and testing datasets. According to Swingler (1996) and Looney (1996), 80% of all data (114 datasets) were considered to develop the models and the remaining 20% of data (28 datasets) were selected to test the constructed model performance. The process of GEP modeling can be summarized in the following five steps:

1. In the first step, the fitness function should be assigned to evaluate the fitness of the generated chromosomes (solutions). In the present study, the root mean square error (RMSE) was considered as the fitness function. The root mean squared error E_i of an individual program i is evaluated by the Eq. (8):

$$RMSE_i = \sqrt{\frac{1}{n} \sum_{j=1}^n (P_{ij} - T_j)^2} \quad (8)$$

where P_{ij} is the value predicted by individual chromosome i for fitness case j (out of n fitness cases), T_j is the real value for fitness case j . Ideally, $P_{ij} = T_j$ and $RMSE_i = 0$. Thus, the RMSE index ranges from 0 to infinity with 0 corresponding to the ideal. As it stands, $RMSE_i$ cannot be used directly as the fitness, since for the fitness proportionate selection, the value of fitness must increase with the efficiency. Thus, for evaluating the fitness f_i of an individual program i ,

the following equation is used:

$$f_i = 1000 \times \frac{1}{1 + E_i} \quad (9)$$

which obviously ranges from 0 to 1000, with 1000 corresponding to the ideal.

2. The second step is choosing the set of terminals and functions to create GEP chromosomes. In this study, terminals are seven input parameters which are listed in Table 3. However, since the choice of the functions does not follow any specific rule, an attempt was made to select the most important functions below for this aim.

Function set = $\{+, -, *, /, \sin, \cos, \tan, \text{Arctan}, \ln, \exp, \text{sqrt}, ^2, ^3, 3Rt\}$

3. The third step is the determination of the chromosomes architecture. The structure of each chromosome depends on the head size and the number of genes. For determining the appropriate values for head size and genes number, a single gene with the head size of 2 was initially set and then increased during each run until the best fit was obtained. Finally, the values of 14 and 5 were obtained as the optimum values for head size and the number of genes per chromosome, respectively. In addition, during this trial and error procedure, the numbers of chromosomes were varied from 30 to 200 and it was found that the population size of 100 gives the best results.
4. In the fourth step, the genetic operators and their rates should be determined. In this study, we used all the genetic operators described in Section 3 and their rates were set according to the literature review and the suggested values by Ferreira (2001) which are appropriate for most of the engineering problems (see Table 3).
5. In the last step, we need to select a function to link the genes (sub-ETs). In the current study, addition (+) was used as a linking function because it gave the better results compared to the other functions such as subtraction, multiplication and division.

By considering all of the above five steps, the GEP model was executed for 2000 generations (iteration) to predict the mill power parameter. The changes related to the fitness value and coefficients of determination (R^2) during the training step are illustrated in Fig. 9. According to this figure, after 370 generations, the fitness function converges to a value of 1000 which is the best fitness value.

The best solution obtained from the GEP modeling is a five genic chromosome in which each one of the genes are linked to each other by the addition (+) function. The expression tree of each gene (sub-ETs) is shown in Fig. 10. Eventually, the mathematical equations belong to each sub-ETs and the general relationships between them are extracted as Eqs. (10)–(14) and (15), respectively. The form of Eq. (15) shows that the GEP algorithm develops a complex nonlinear expression

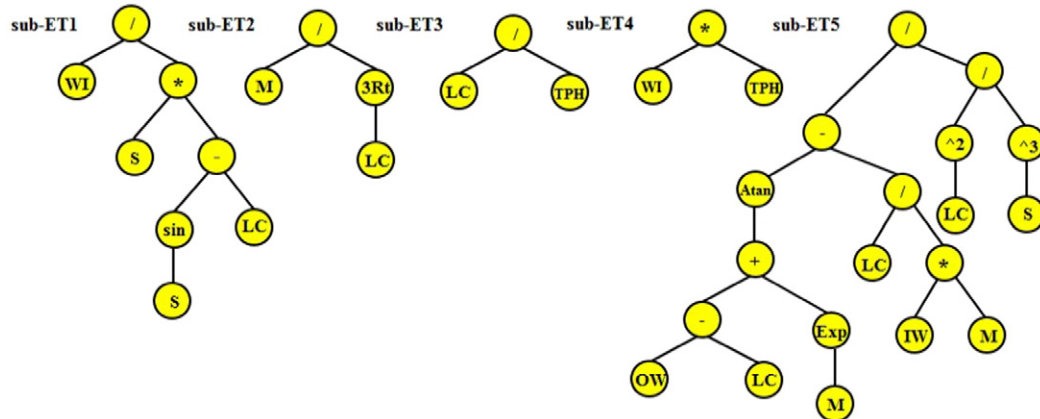


Fig. 10. The sub-ETs of the five genic chromosomes for the proposed GEP model.

to accurately predict the mill power which is too difficult to obtain via the traditional regression-based techniques.

$$\text{sub-ET1} = \frac{WI}{S \times (\sin(S) - LC)} \quad (10)$$

$$\text{sub-ET2} = \frac{M}{\sqrt[3]{LC}} \quad (11)$$

$$\text{sub-ET3} = \frac{LC}{TPH} \quad (12)$$

$$\text{sub-ET4} = WI \times TPH \quad (13)$$

$$\text{sub-ET5} = \frac{\text{Atan}(OW - LC + \text{Exp}(M)) - \frac{LC}{IW \times M}}{\frac{LC^2}{S^3}} \quad (14)$$

$$\text{Mill power} = \frac{WI}{S \times (\sin(S) - LC)} + \frac{M}{\sqrt[3]{LC}} + \frac{LC}{TPH} + (WI \times TPH) + \frac{\text{Atan}(OW - LC + \text{Exp}(M)) - \frac{LC}{IW \times M}}{\frac{LC^2}{S^3}} \quad (15)$$

4.3. Model performance evaluation

In this section, the performance prediction of the two developed predictive models namely GEP and NLMR for the prediction of SAG mill power is evaluated. For this aim, three statistical performance indices including coefficient of determination (R^2), root mean square error (RMSE) and variance account for (VAF) were used. The mathematical equations of these indices are expressed as follows:

$$R^2 = 1 - \frac{\sum_{i=1}^N (x_{imes} - x_{ipred})^2}{\sum_{i=1}^N (x_{imes} - \bar{x})^2} \quad (16)$$

$$\text{RMSE} = \sqrt{\frac{1}{N} \sum_{i=1}^n (x_{imes} - x_{ipred})^2} \quad (17)$$

$$\text{VAF} = \left[1 - \frac{\text{var}(x_{imes} - x_{ipred})}{\text{var}(x_{imes})} \right] \times 100 \quad (18)$$

where $\bar{x}$ is the mean of the measured values, N is the total number of dataset and x_{imes} and x_{ipred} are the measured and predicted values by models, respectively. Theoretically, in the best conditions, R^2 is one, VAF is 100 and RMSE is zero. The calculated values of performance indices are listed in Table 4. The obtained values of 0.9967, 47.3310 and 99.3539 for R^2 , RMSE and VAF in training datasets, and also 0.9946, 42.2343 and 99.2912 for testing datasets, respectively indicate the relative superiority of the constructed GEP model in predicting the SAG mill power to NLMR model. Figs. 11 and 12 show the measured values of mill power in comparison to the predictions of GEP and NLMR models in the testing process, respectively. According to these figures, the obtained results for mill power by the GEP model are closer to the measured mill powers compared to the NLMR model.

Table 4
Obtained performance indices values for developed models.

Developed model	Training			Testing		
	R^2	RMSE	VAF	R^2	RMSE	VAF
GEP	0.9967	47.3310	99.3539	0.9946	42.2343	99.2912
NLNR	0.7714	71.272	76.815	0.5788	68.396	53.395

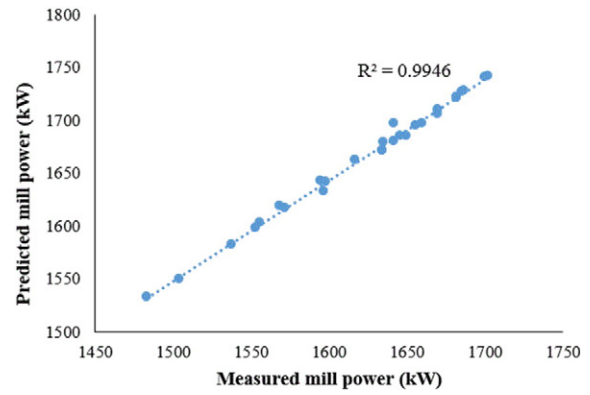


Fig. 11. Predicted mill power by GEP in the testing process vs. actual measurement.

4.4. Sensitivity analysis

The most effective input parameter on output can be evaluated using the sensitivity analysis. In this regard, the sensitivity of the mill power model that is developed using GEP is calculated based on the fact that each parameter changes between the minimum and maximum of its value while the other parameters remain constant at their average value. Thus, it is possible to investigate the effect of each change in input parameter on the output parameter (mill power). For example, for the investigation of the TPH changing sensitivity on the mill power, all parameters except TPH are constant in their average value while the TPH changes. Therefore, sensitivity to all input parameters was similarly evaluated. The units of each input parameter to estimate mill power are different. The simultaneous display of input parameter changes on mill power is difficult because of the difference in size and value of the input parameters. Therefore, the parameters (between their minimum and maximum value) that were used for sensitivity analysis should be standardized. The z-score is used for standardization of parameters.

$$Z = \frac{(X_i - \bar{X})}{\sigma} \quad (19)$$

where, X_i , $\bar{X}$, σ and Z are the i th variable, the average and standard deviation of input parameters and the parameter standardized value, respectively.

The simultaneous display of mill power changes versus changes any input parameters are shown in Fig. 13, which shows that the mill power is more sensitive to TPH and WI than the other input parameters.

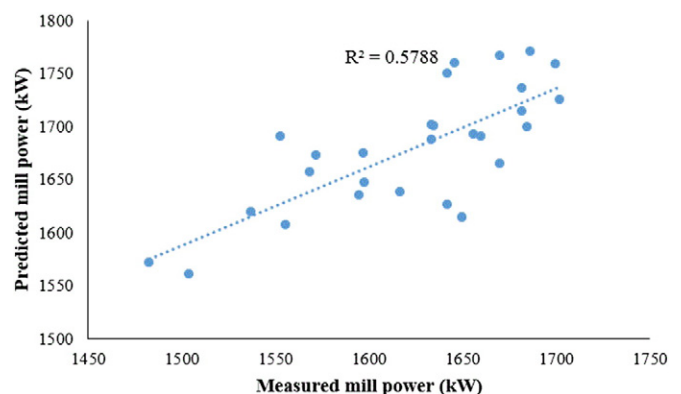


Fig. 12. Predicted mill power by NLMR in the testing process vs. actual measurement.

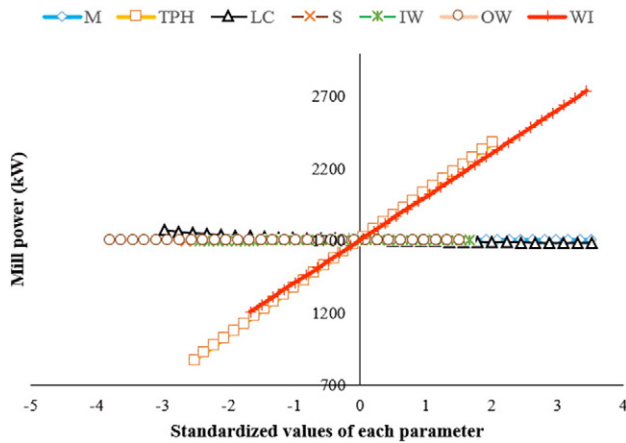


Fig. 13. Sensitivity analysis of input parameters on the mill power model.

5. Conclusion

Performance prediction of semi-autogenous (SAG) mills in the full-scale mill performance can be evaluated for circuit optimization by suitable modeling and simulation. Mill powers of the semi-autogenous mill have an effective influence on the mill performance. In this study, the mill power model was developed for the full-scale SAG mill using GEP and MLMR methods. The influence of effective parameters such as feed moisture, mass flowrate, mill load cell weight, SAG mill solid percentage, inlet and outlet water to the SAG mill and work index are evaluated to predict the SAG mill power in full-scale. The efficiencies of GEP and NLMR were compared with each other for the prediction of the SAG mill power. The results showed that the efficiency of the new proposed GEP model is more than that of the NLMR in predicting the SAG mill power. The correlation coefficient and mean square error values for the testing sets for SAG mill power using the GEP and NLMR methods were 0.9946, 42.2343 and 0.5788, 68.396, respectively. The correlation coefficient and the mean square error values of the testing sets in GEP are less than those in NLMR that show the greater efficiency of GEP for this study. The sensitivity analysis was also evaluated to determine the most effective input parameters on output. It shows that the mill power is more sensitive to TPH and WI than other input parameters. The results in this study showed that the proposed model, according to the parameters affecting the SAG mill power, can be used to predict the SAG mill power with a reasonable error.

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